



Figure 1: Network Simulator 2 architecture

Supporting tools with NS2

Let's explore the supporting tools that come with the NS2 installation, which help set up your environment and run your simulations.

NAM (Network AniMator): NAM is a TCL-based animation tool for viewing network simulation traces and real traces of packet data. To use NAM, you have to first generate a trace file that contains topological information like nodes, links and packet traces.

Once the trace file is generated, NAM will read it, create a topology, pop up a window, do the layout if necessary, and then pause at the time of the first packet in the trace file. NAM provides control over many aspects of animation through its user interface, and does animation using the following building blocks: node, link, queue, packet, agent and monitor. The official NAM user manual can be found at <http://www.isi.edu/nsnam/nam/>

XGraph: XGraph is a plotting program that is used to create graphic representations of simulation results. It is important because it allows some basic animation of data sets. The animation only pages through data sets in the order in which they are loaded. It is quite crude, but useful if all the data sets are in one file in the time order, and are put out at uniform intervals. Also, the code will take derivatives of your data numerically and display these in a new XGraph window.

NS2 functionalities

Wired world

Routing: distance vector (DV), link state (LS), and multicast Transport protocols: TCP, UDP, RTP and SCTP
Traffic sources: WEB, FTP, telnet, CBR, and Stochastic
Queueing disciplines: Drop-Tail, RED, FQ, SFQ, and DRR
QoS: IntServ and Diffserv emulation

Wireless

Ad hoc routing (AODV, DSDV) and mobile IP
Directed diffusion, and sensor-MAC

Installing NS2 on Ubuntu

- Download the latest release of the ns-allinone package from the official NS2 weblink <http://www.isi.edu/nsnam/ns/ns-build.html>

- Open the terminal and change the working directory to where you downloaded the ns-allinone package.
- Now untar (uncompress) the package using the following command:

```
tar xzvf filename
```

Change the directory to ns-allinone2.29, run the installation script using the following command and wait until the installation is successfully completed:

```
./install
```

- After successfully installing the NS2 package, configure the `.bashrc` file, which is present in the following path:

```
home/ns2username
```

Edit the `.bashrc` file using the following command:

```
gedit .bashrc
```

Set the following path in the last line of `.bash`

```
# export PATH="$PATH:/home/ns2username/ns-allinone-2.29/bin:/home
/home
/home/ns2username/ns-allinone-2.29/tcl8.4.11/unix:/home/
ns2username/ns-allinone-2.29/tk8.4.11/unix"
export LD_LIBRARY_PATH="/home/ns2username/ns-allinone-2.29/
otcl-1.11,/home/ns2username/ns-alli none-2.29/lib"
export TCL_LIBRARY="/home/ns2username/ns-allinone-2.29/
tcl8.4.11/library"
```

- If the path set is correct, open the new terminal and run the following in the home directory:

```
ns/
```

If you see a % sign displayed in the terminal, congratulations, you have installed NS2 successfully!

- Now to validate the installation, run the following command (optional):

```
cd /ns-allinone-2.29/ns-2.29 / ./validate
```

Execution of a simple TCL file in NS2

In this section, let's cover the execution of TCL files for a simple network architecture.

```
set ns [new Simulator] //This creates an NS2 simulator object
```

```
set tracefile [open out.tr w]
```

```
$ns trace-all $tracefile //Open trace file
```